Lab 8: Parking Garage Management System

This lab walks you through the analysis, design and implementation of a small parking garage system. This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer.

Part A: Requirements analysis

This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer. Identify the actors that interact with the garage system. For each actor, describe at least two goals in one sentence each.

Part B: Domain model

This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer. Your domain model must cover at least the following concepts:

- Levels and parking spots of different sizes
- Vehicles and their size classes
- Tickets with entry and exit times
- Payments for a ticket

Part C: Implementation

This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer. Implement the domain model in an object-oriented language of your choice. At minimum, your program needs to support:

1. Assigning the smallest free spot that fits a vehicle
2. Issuing a ticket when a vehicle enters

3. Calculating the fee from the entry and exit times
4. Freeing the spot when the vehicle leaves

The fee rules should be easy to change without editing the ticket class.

Part D: Command-line interface

This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer. Your command-line interface should support these operations:

- Park a vehicle
- Remove a vehicle and print its fee
- Show the free spots on each level

Part E: Testing and submission

This part builds on the previous one, so keep your earlier notes at hand and reuse the names you chose. Read the whole part before you start writing anything. The garage in this lab is a fictional multi-level building next to the campus library; it has three levels, and each level has spots for motorcycles, compact cars and large cars. Drivers take a ticket at the entrance barrier and pay at the exit barrier before leaving. The examples in the lecture slides use a simpler garage with only one level, which is a good starting point for your own thinking. Discuss your ideas with your lab partner, sketch them on paper first, and only then move on to the computer. Write unit tests for the spot assignment and the fee calculation. Use only the standard library of your language.

- Submit the source code as a ZIP file.
- Submit the class diagram as a PDF.

Criterion	Marks
Requirements analysis	4
Domain model	6
Implementation	10
Command-line interface	4
Tests	6

Total: 30 marks